

Image Segmentation for Pavement Crack Detection System

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Abstract— Pavement distress refers to the condition of pavement surface in terms of its general appearance. Cracks is a type of pavement distress and commonly occur in a road infrastructure. Crack on a pavement surface shows an early sign of pavement problems and aging. Thus, it is important to detect a serious crack as soon as possible to avoid any road accident that might occur. This study shows a comparison of three popular methods of image segmentation; watershed, k-means clustering and Otsu thresholding for pavement crack detection system in terms of its overall performance. Sample of crack images from three different types of crack such as transverse, longitudinal and crocodile crack are captured manually using digital camera and from online sources. The image is then imported into MATLAB software where it will be compressed but without reducing its quality and pixels intensity. The compressed image is then converted into grayscale to make it easier for analyzing as the system only need to work with one layer instead of three layers (RGB). The contrast of the image is then stretched to increase the level of contrast between the crack and the background. Then, the image will be segmented using three different segmentation method that are mentioned above. Lastly, morphological operation is used to reduce the noise from the image segmented. The result of the segmented image will be analyzed in term of its Structural Similarity Index (SSIM) and Mean Squared Error (MSE). The performance of the system is measure using images with a high level of contrast between the crack and the surface and images with a low level of contrast between the crack and the surface.

I. INTRODUCTION

Road network plays an important and crucial role in today's world of globalization. Road is necessary for economic activity as most of goods and materials are transported on land using lorry, van, car, motorcycle and other methods of transportation. Likewise transportation services such as taxi and bus. With a lot of large and heavy vehicles using road as a medium to bring goods and passengers, it causes stress to the pavement surface and bring damage to the road infrastructure over time [1]. Pavement distress refers to the condition of a pavement surface in terms of its general look [2]. A distressed pavement may be fractured, distorted or disintegrated. Pavement distress is usually caused by weathering, aging, and stress caused by heavy load [3]. There are many types of pavement distress such

as surface defects, surface deformation, cracks, patches and potholes [3]. This study will focus on crack type pavement distress that are transverse, longitudinal and crocodile cracks. Image segmentation is a crucial step in image processing to extract important features in an image. Image segmentation divides an image into multiple areas, generally based on pixel intensity, gray scale values, spatial characteristics and texture characteristics [4]. Segmentation techniques are intended to define and separate regions of interest within the image [5]. Currently, observation and evaluation of road pavement distress are done manually by the worker with different individuals comes out with different kind of result [1]. With a vast road network connecting one place to the other place or one state to the other state, maintaining the road infrastructure is a difficult work. An advanced monitoring system with an automatic image processing should be implemented with a rapid development of road network system to ensure the quality of the road infrastructure.

A lot of research for pavement crack detection have been made in recent years. A method based on multiple scale and differential box dimension is introduced recently [1]. It shrinks the image by analyzing the crack direction. It claimed to be effective in removing noise while still maintaining the crack information, and then use the logarithmic transform for image enhancement, and later apply for the improved differential box-counting method for image segmentation. Another method is based on analyzing the difference between gray neighbors, analysis of the local minimum gray and the sub-block label. This joint automatic detection method of the pavement distress is proposed in 2016 by Dang et al. [6]. Also in 2016, Zhang et al. proposed a new explanation of the cracks based on the spatially clustered pixels with similar grey level [7]. A distress region that contains cracks can be labeled with the defining of some credibility factor that indicates the reliability of that region with the help of a new concept called Region of Belief (ROB) that aid with the following detection that was introduced. Such ROB is extracted using a new algorithm that has been devised. An idea of a new region growing algorithm is also proposed for crack detection, which features starting with an ROB seed, determining the searching scope with a specially devised rule, and searching and merging a ROB with different regions following a similarity criterion

which synthetically takes different cues into consideration. The limitation of this method is that if the intensity between the crack pixels and background pixels is low, it would be difficult for the program to differentiate between those pixels as the key to their projects is the grey-level difference between crack pixels and background pixels. In 2014, Henrique and Paulo developed an image processing toolbox called CrackIT [8]. The CrackNet, is an architecture based on the Convolutional Neural Network (CNN) proposed by Zhang for automated pavement crack detection on 3D asphalt surfaces with the primary objective of pixel-perfect accuracy [9]. Other methods such as thresholding was proposed to find cracks by setting global or local thresholds value and comparing it with pixel value intensity of the image with various limitations for images with complex illuminance [10]. Filter-based algorithms was developed to find cracks of anticipated responses with limitations in detecting cracks of weak responses to predesigned filters [11]. The CrackTree introduced in 2012 demonstrated a strong capability of detecting discontinued cracks but without considering the actual crack width [12].

The objectives of this project is to develop image segmentation algorithm for pavement crack images based on three popular segmentation technique that is watershed, k-means clustering, and Otsu thresholding. Then, to analyze the performance of each method.

II. METHODOLOGY

This section describes the methodology for pavement crack image segmentation. Fig. 1 shows the process flowchart for the image segmentation technique applied.

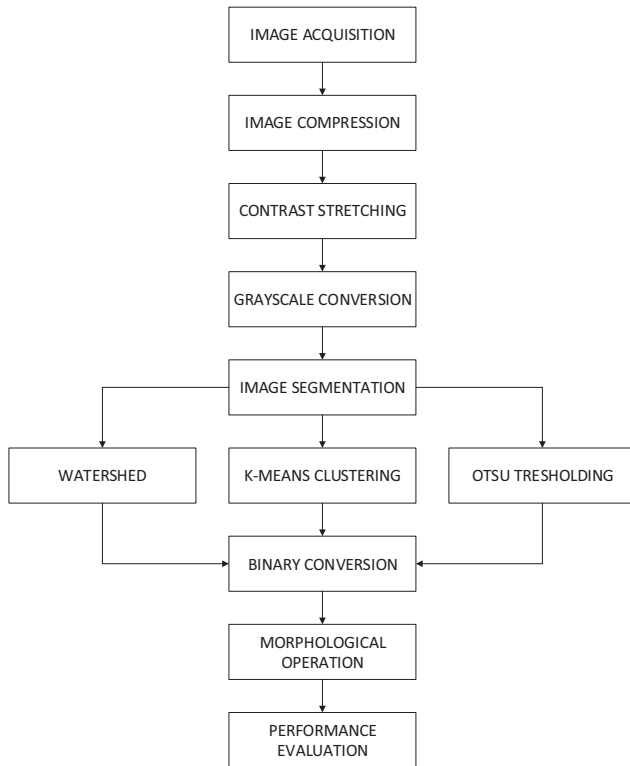


Figure 1: Process flowchart

A. Image acquisition

Sample images from the internet and manual capturing of the image using digital camera are used in this study.

B. Image compression

Usually a high precision camera is used to capture the pavement image. The resolution of the image can be up to 2048×2048 pixels, and even up to 4096×4096 pixels. If the image size is too large, the computer processing speed will be slowed. Hence, the size of the image need to be reduced first while maintaining the image quality and reduce the number of unnecessary pixels. When resizing an image, the ratio of the image need to be maintained to keep the image from getting distorted.

Width Formula:

$$\text{original height} / \text{original width} * \text{new width} = \text{new height}$$

Height Formula:

$$\text{original width} / \text{original height} * \text{new height} = \text{new width}$$

C. Contrast stretching

Contrast stretching or often called as normalization is a simple image enhancement technique that attempts to improve the contrast in an image by increasing the range of intensity values it contains to span a desired range of values, e.g. the the full range of pixel values that the image type concerned allows. The upper and lower pixel value limit are specified over which the image is to be normalized. Then, the images is scanned to find the highest and lowest pixel values currently present in the image labelled as c and d . Then, each pixel, P is scaled using the following function:

$$P_{out} = (P_{in} - c) \left(\frac{b-a}{d-a} \right) + a \quad (1)$$

D. Grayscale conversion

The contrast stretched image is converted into grayscale image with a pixel value range from 0 to 255. It is easier to work with one layer of color instead of three layers that is RGB. It is the process of converting RGB values (24-bit) into grayscale values (8-bit) by forming a weighted sum of the R, G, and B components:

$$0.289 * R + 0.587 * G + 0.1140 * B \quad (2)$$

E. Image Segmentation

Three popular image segmentation method are used in this study to analyze the performance of each method.

- **Watershed**

An image is treat like a surface with bright area is consider as a high value and dark area as a low value creating a term called watershed line and catchment basin. It treats the

image it operates upon like a topographic map, with the brightness of each point representing its height, and finds the lines that run along the tops of ridges.



(a)



(b)

Figure 2: (a)Original image & (b)after applying watershed segmentation

- K-means clustering

K-Means is a least-squares subdivide method that separate a group of target into K groups. It aims to partition “n” observations into “k” clusters in which each observation belongs to the cluster with the nearest mean.



(a)



(b)

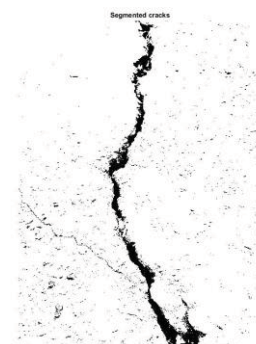
Figure 3: (a)Original image & (b)after applying k-means clustering segmentation

- Otsu thresholding

Thresholding is the conversion of the pixel intensity level into binary value, 1 & 0 that represent either white or black. This can be done by comparing the intensity value of each pixel with an already set threshold value. For example, if the pixel value is higher than the threshold value, it is set as white with a value of 1. Same goes if the pixel value is lower than the threshold value, it is set as black with a value of 0.



(a)



(b)

Figure 4: (a)Original image & (b)After applying Otsu thresholding segmentation

F. Binary conversion

A grayscale image is converted into binary image based on threshold value by replacing all pixels in the input image with luminance greater than level with the value 1 (white) and replacing all other pixels with the value 0 (black).

G. Morphological operation

Morphological operation function in Matlab such as *bwareaopen* is used to remove small objects (noise) in a binary image. It removes all connected components (objects) that have fewer than P pixels from the original binary image leaving only the crack line. Then, producing another binary image. This method is also called area opening. However, image with very low contrast level is difficult to remove the noise as the pixel value of the noise is the same as the pixel value of the crack line.

III. RESULTS AND DISCUSSION

10 images of cracks with three different types of cracks and different contrast level are tested for the three segmentation methods. Fig. 2 shows the result of segmentation of a sample crack image using watershed, k-means clustering and Otsu thresholding as compare to the groundtruth image. Fig. 5 shows the overall performance of the three segmentation methods.

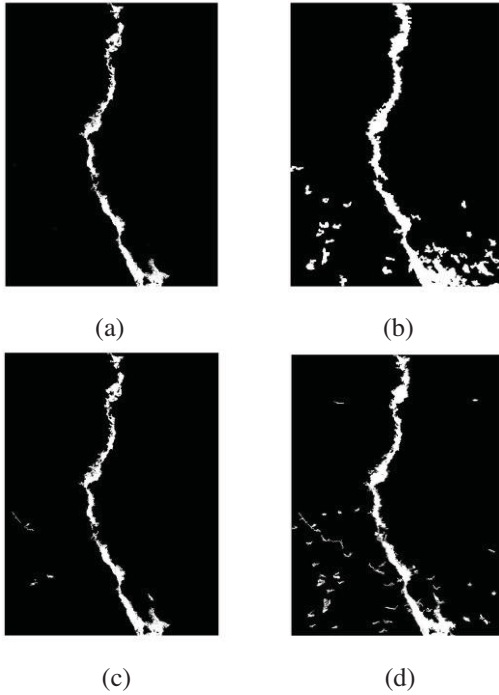


Figure 5: (a) Groundtruth, (b) k-means clustering, (c) Otsu and (d) Watershed image segmentation.

Table 1: Overall performance in SSIM

Image	Watershed	K-means	Otsu
1	98.62%	98.08%	99.84%
2	95.52%	93.72%	99.99%
3	97.25%	94.08%	99.73%
4	99.84%	97.20%	99.31%
5	98.44%	91.57%	99.24%
6	99.97%	75.34%	99.07%
7	99.17%	68.14%	96.52%
8	93.35%	91.59%	99.57%
9	88.50%	86.82%	90.86%
10	81.40%	78.05%	97.45%
Average	95.21%	87.46%	98.16%

Table 2: Overall performance in MSE

Image	Watershed	K-means	Otsu
1	177.3294	285.0328	26.6090
2	604.8418	1291.1695	0.0000
3	245.1113	984.9271	27.0109
4	20.5223	456.9744	62.5272
5	149.8565	1145.2838	47.3984
6	1.6223	7049.5654	96.9857
7	76.4656	6776.4117	330.7829
8	785.1860	1774.2817	34.7379
9	1282.4245	2306.9060	1011.6132
10	2712.9713	4717.9233	428.8667
Average	605.6331	2678.8476	206.6532

The accuracy test is done using Structural Similarity Index (SSIM) and Mean Squared Error (MSE). SSIM is used to measure the similarity between two images. It is calculated on various windows of an image. The measure between two windows x and y of common size $N \times N$ is

$$SSIM(x, y) = \frac{(2\mu_x + \mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (3)$$

The SSIM index satisfies the condition of symmetry $SSIM(x, y) = SSIM(y, x)$. The higher the percentage of similarity between the segmented image and groundtruth image, the higher the accuracy of the segmented method.

Mean Squared Error (MSE) measure the absolute error of the segmented image when compared to the groundtruth image. It is defined as

$$MSE = \frac{1}{N \times M} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} [X(i, j) - Y(i, j)]^2 \quad (4)$$

The more similar Y to X, the smaller the MSE. The greatest similarity is achieved when the MSE is equal to zero.

Based on the result obtained, Otsu thresholding method gave the highest percentage in terms of structural similarity index and mean squared error as compared to the other two. Otsu thresholding perform better in image with a high level of contrast between the crack line and the road surface. However, it cannot join similar discontinued crack line as it only convert the pixel value into gray level. K-means clustering perform better if the crack line is uniform in shape as it tends to find the centroid between two cluster that are crack and background. If there is a lot of noise in the image, it would be difficult to find the centroid and thus making it less reliable. But it can join discontinue crack line unlike Otsu as it find the centroid between two cluster. If the discontinued crack line pixel value is inside the crack cluster, then it belongs to the crack line and join the discontinued crack line together. K-means clustering give the lowest performance especially in image with a lot of noise. Meanwhile, watershed tend to work better in an image with a lot of colors especially if the crack line is visible and vivid.

IV. CONCLUSION

This study compared the performance of the three popular segmentation methods that are watershed, k-means clustering and Otsu thresholding for pavement crack detection system. From the analysis, Otsu method based on gray level thresholding still outperform the other two segmentation method in crack detection. The ability to segment even a non-uniform crack makes it one of the popular approach in crack segmentation. Thresholding is also widely used in image processing. For watershed, even though it is one of the popular method in image segmentation, it has never been used in crack detection. It is commonly used to separate touching object in an image. This is the first study to ever use watershed transform to segment a crack. Based on the analysis, it is almost on par with Otsu thresholding segmentation performance. Hence, it can be improved further to make it more reliable and efficient. In k-means clustering case, it showed that it is not the suitable method for crack detection. An improved algorithm has been made to counter the non-uniform crack problem but it is still not practical as it can only work with single crack [13].

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